CASE STUDY: INTELLIGENT SYSTEMS IN PRODUCTION

REVISED TOPIC

# AOI Decision Gate

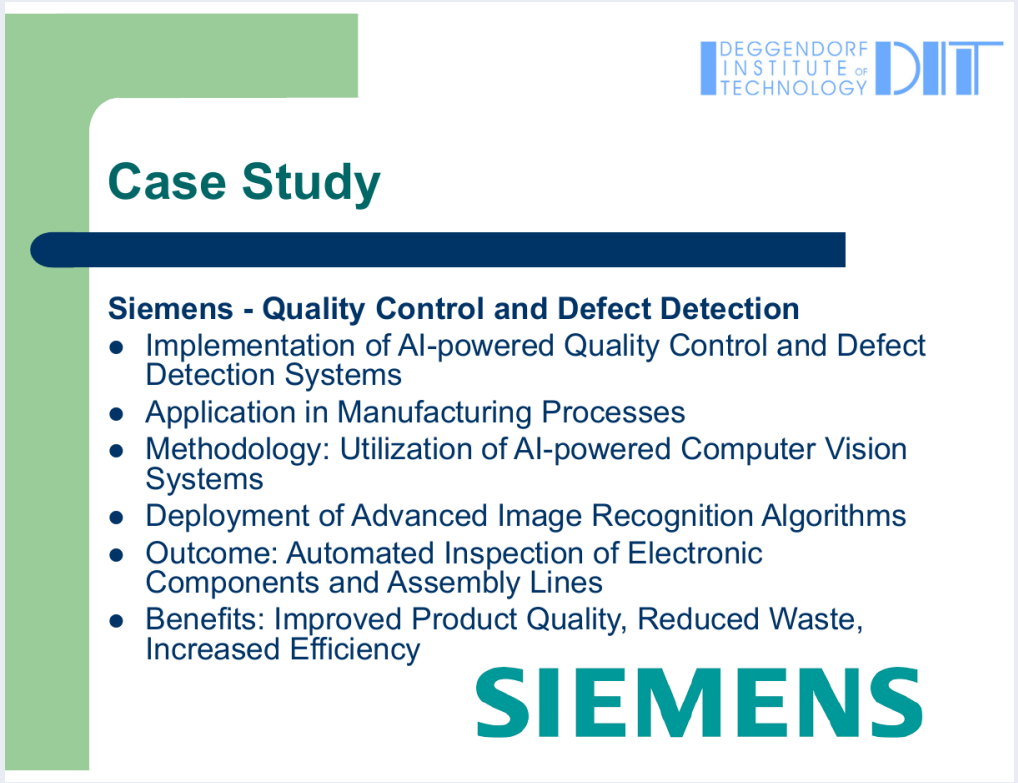
AI decision gate for false-call reduction in automated optical inspection of electronics assembly

Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas

Team 1 · Guided by Dr. Hamidreza Heidari · 23 October 2026 · github.com/bodhi584/aoi-decision-gate

WHERE THE TOPIC COMES FROM

# Based on the lecture example: Siemens, p. 18




| Lecture slide | Our project |
|---|---|
| AI-powered quality control and defect detection | **AI decides: real defect or false call** |
| Application in manufacturing processes | **SMT electronics assembly, after soldering** |
| AI-powered computer vision systems | **Works on the AOI camera system's measurements** |
| Advanced image recognition algorithms | **Classifier on AOI image measurements** |
| Automated inspection of electronic components | **Real Siemens line data (Bavaria)** |
| Improved quality, reduced waste, efficiency | **Fewer manual checks, no extra defects released** |

TECHNISCHE HOCHSCHULE DEGGENDORF THD

THE INDUSTRIAL PROBLEM

# Too many false alarms after optical inspection, today

**98 %**

of the boards flagged by the AOI machine are fine (false calls) in the Siemens line data.

Every one goes to a manual inspection station: “a major manual labour effort” (Pfab et al. 2024, Siemens).

### 21 Oct 2026 · Siemens and NVIDIA, Erlangen electronics factory

AI second check of AOI results: is it a real defect or a false call?

Project targets: about 40 % fewer false calls, reduced manual verification.

### 19 Oct 2026 · Siemens, Erlangen

Factories in high-wage countries need 7–9 % more productivity every year.

Siemens invests €500 million in Erlangen as a showcase for AI in manufacturing.

Sources: Siemens press release, 21 Oct 2026; technology.org, 20 Oct 2026; Pfab et al. 2024, Data in Brief 53:110110

TECHNISCHE HOCHSCHULE DEGGENDORF THD

KEY FACTORS

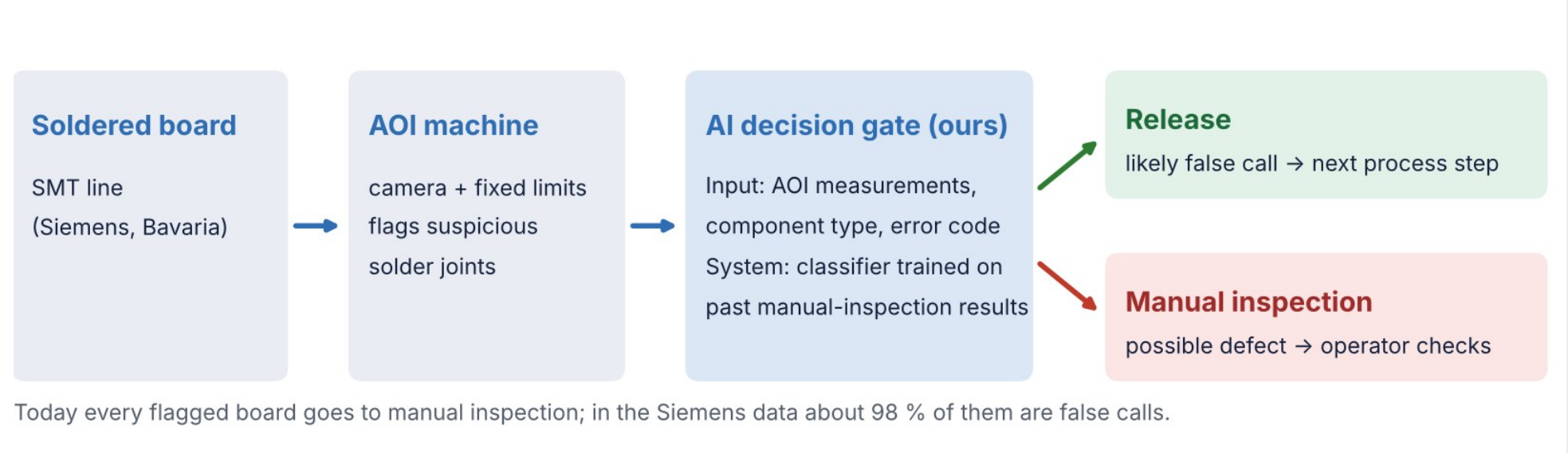
# What decides whether the gate helps the line

| | |
|---|---|
| **False-call rate** | Each false call costs operator time at the manual station |
| **Missed defects** | A released defective board reaches the customer: must stay ≤ 1 % |
| **Drift over time** | Materials, settings and products change; last month’s model may not fit |
| **Class imbalance** | Only about 1 % of flagged joints are real defects |
| **Label quality** | Labels come from operators and may contain errors |

TECHNISCHE HOCHSCHULE DEGGENDORF

INPUT → SYSTEM → OUTPUT

# Where the gate sits on the line




## Input

AOI measurements of each flagged joint (~70 values), inspection type, component type, error code

## System

Classifier trained on past manual-inspection results + release threshold (≤ 1 % defective boards released)

## Output → impact

Release or send to manual inspection → less manual work, same quality

TECHNISCHE HOCHSCHULE DEGGENDORF THD

METHODOLOGY

# CRISP-DM, evaluated by a replay simulation of the line

| 1 · Business understanding | 2 · Data understanding | 3 · Data preparation | 4 · Modelling | 5 · Evaluation | 6 · Deployment |
|---|---|---|---|---|---|
| false-call problem, KPIs, limits | structure, imbalance, drift | joints → boards, strict time order | gradient-boosted trees; neural network | replay simulation, production KPIs | replay demo, run instructions |

## Replay simulation (our “baseline simulation”)

The 132 recorded days are played back in time order. Each day the gate decides every flagged board with a model trained only on earlier days, exactly as on the real line.

We compare update policies (train once, retrain weekly on all past days, retrain on recent days) and measure released false calls, released defects and saved inspections.

Why not a random split? It mixes future and past and overstates what a factory gets (next slides).

TECHNISCHE HOCHSCHULE DEGGENDORF

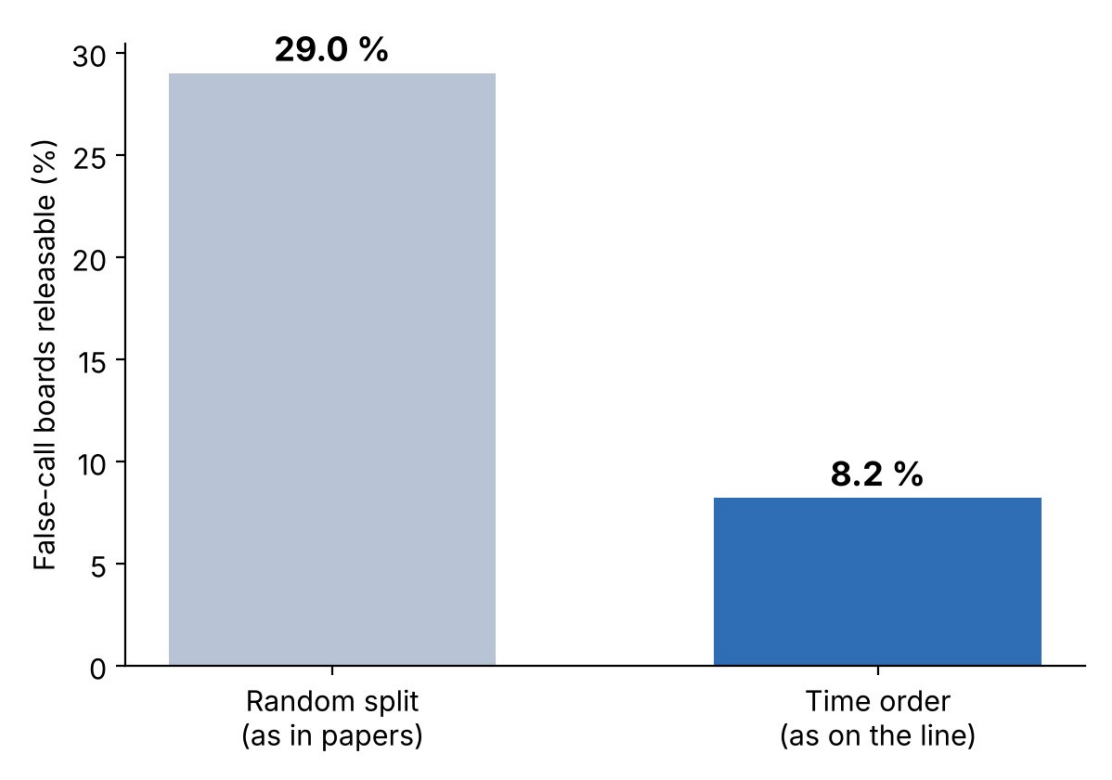
DATA

# Real production data published by Siemens

| | |
|---|---|
| **Source** | Pfab, Eichler, Mallandur, Rothering (2024), Siemens AG, Data in Brief 53, 110110 |
| **Line** | One SMT production line of Siemens AG in Bavaria, 132 days |
| **Size** | 440,274 flagged solder joints on 39,742 boards; 5 inspection types |
| **Features** | ~70 AOI measurements + component type, error code, angle (anonymised, scaled 0–1) |
| **Label** | Manual inspection result: false call or true defect (875 defective boards) |
| **Drift** | Early vs. late days separable from measurements alone: AUROC 0.966 |
| **Licence** | CC BY-NC 3.0 (Siemens AG): downloaded by script, not redistributed |

TECHNISCHE HOCHSCHULE DEGGENDORF THD

# Time order is much harder than a random split




## What it shows

Gradient-boosted trees, trained on the first 70 % of records.

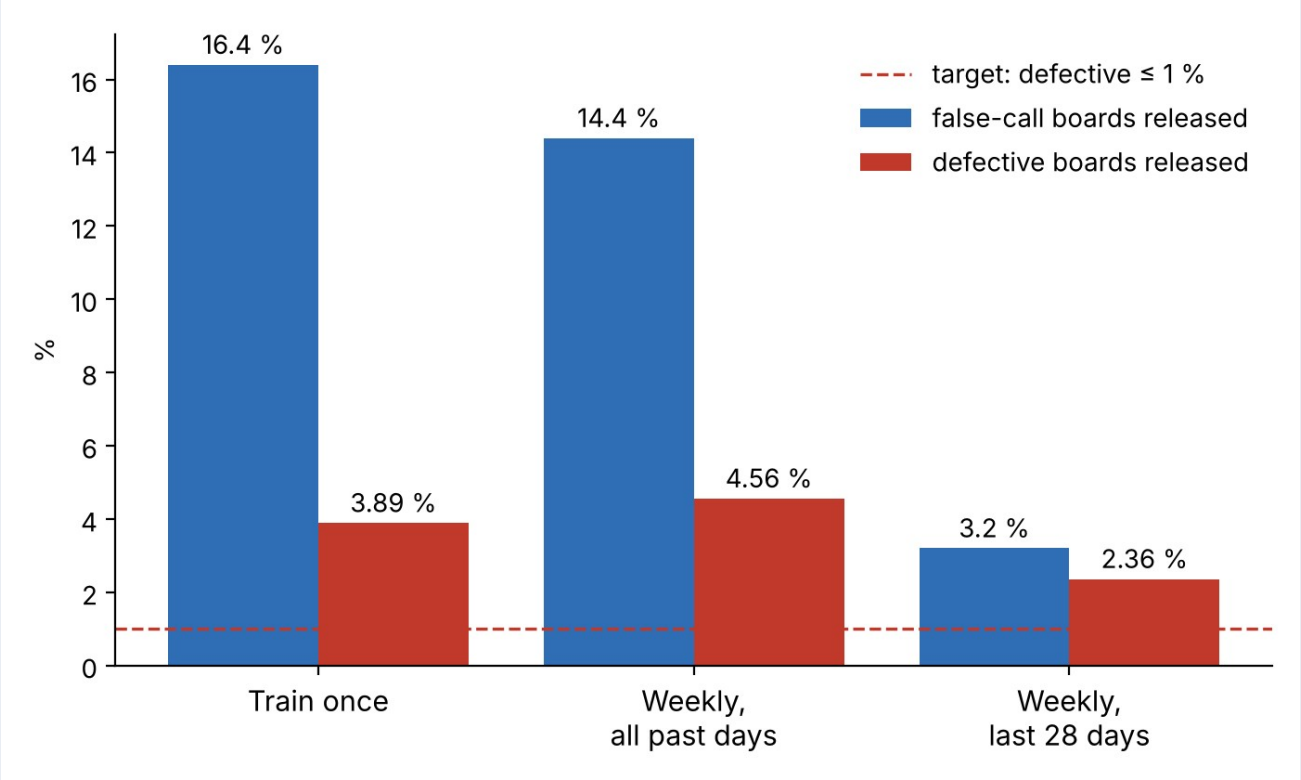
Share of false-call boards that could be released while 99 % of defective boards still go to manual inspection.

Random split: 29 %, AUROC 0.966. Time order: 8.2 %, AUROC 0.889.

The line changes over time: drift is the real challenge, and papers with random splits overstate results.

TECHNISCHE HOCHSCHULE DEGGENDORF

# Thresholds from past days do not hold on later days




## What it shows

39 production days, 17,366 flagged boards replayed.

Up to 16 % of false-call boards released (2,750 manual inspections saved).

But 2.4–4.6 % of defective boards released: the 1 % limit is broken by drift.

→ Core of the project: drift detection and update policy.

GOALS AND KPIS

# Fixed now, before the main experiments

| | Baseline today | Target (12 Jan) |
|---|---|---|
| **Defective boards released** | 2.4–4.6 % | **≤ 1 %** |
| **False-call boards released (inspections saved)** | 8.2 % | **≥ 25 %** |
| **Drift alarm** | none | **alarm when performance drops** |
| **Model comparison** | trees only | **trees vs. neural network** |

TECHNISCHE HOCHSCHULE DEGGENDORF

# What the project covers, and what it does not

## Conditions

One Siemens SMT line, five inspection types, 132 days

Gate placed after the AOI, before manual inspection

Decisions per board; at most 1 % defective boards released

Public data only, no personal data

## Limitations

Measurements anonymised: no physical names

Labels from operators may contain errors

No images; no other lines or plants

Licence non-commercial (course use only)

Risk: drift limits the release rate → fallback: report what works and why

# Everything on GitHub, reproducible

## Tech stack

Python · pandas · scikit-learn (gradient-boosted trees)

PyTorch (neural network, from P3)

pytest + GitHub Actions: tests on every push

Neural networks (lecture p. 12, 17) · big data processing (p. 10)

No large language model

## Deliverables

github.com/bodhi584/aoi-decision-gate

Decision gate + replay simulation (code, tests, CI)

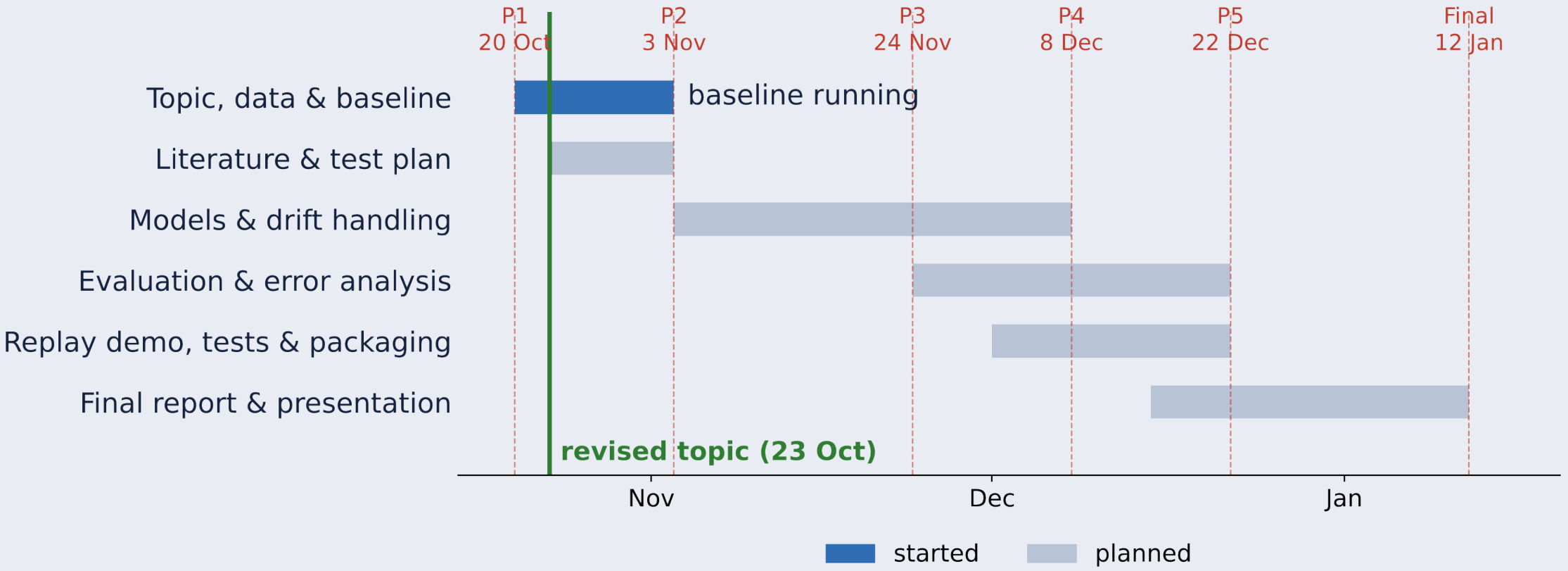
Result files for every number we report

Replay demo + demo video (P5)

Final report and presentation

TIMELINE

# Milestones follow the Software & Simulation track (p. 33)




TECHNISCHE HOCHSCHULE DEGGENDORF

REFERENCES

# Sources


Pfab, K., Eichler, R., Mallandur, A., Rothering, M. (2024). Data of automated optical inspection of surface-mounted technology electronic production. Data in Brief 53, 110110.

Siemens (21 Oct 2026). Siemens and NVIDIA advance AI-powered PCB inspection from digital twin to factory. Press release.

technology.org (20 Oct 2026). Siemens aims to raise factory automation to 80 % with AI.

Albayari, J., Patra, P., Patel, D., Santos, D. (2025). Enhancing false call detection in automated optical inspection using XGBoost classifier and XAI algorithms. FAIM 2025, Springer.

Schmitt, J., Bönig, J., Borggräfe, T., Beitinger, G., Deuse, J. (2020). Predictive model-based quality inspection using machine learning and edge cloud computing. Advanced Engineering Informatics 45, 101101.

Chapman, P. et al. (2000). CRISP-DM 1.0: Step-by-step data mining guide. SPSS.

Lecture slides, Case Study Intelligent Systems in Production, p. 10, 12, 17, 18, 33.

# Thank you

github.com/bodhi584/aoi-decision-gate

Team 1 · Yu Han · Melisa Subasi · Meron Hagos · Alejandro Rivas